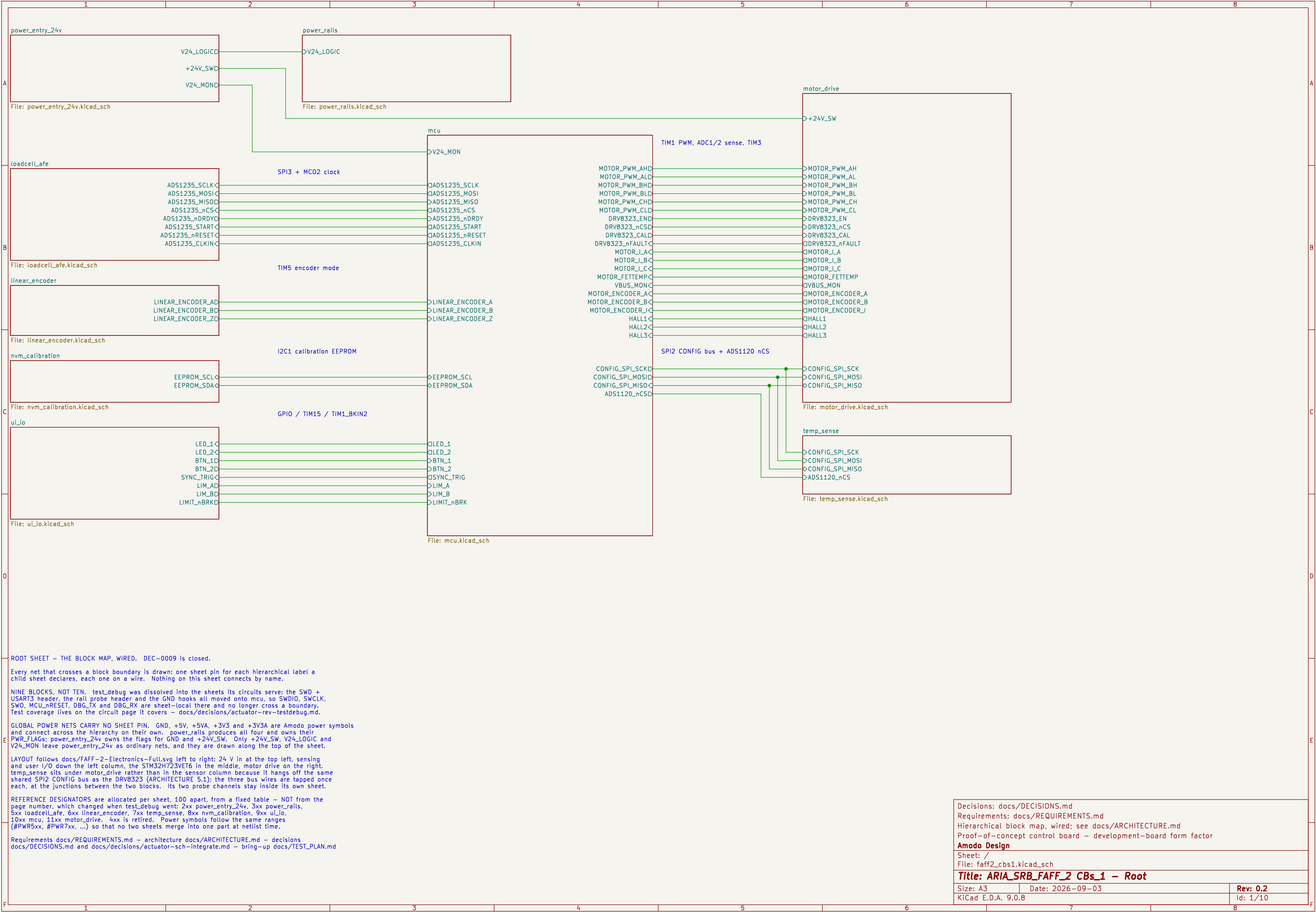




ROOT SHEET – THE BLOCK MAP, WIRED. DEC–0009 is closed.

Every net that crosses a block boundary is drawn: one sheet pin for each hierarchical label a child sheet declares, each one on a wire. Nothing on this sheet connects by name.

NINE BLOCKS, NOT TEN. test\_debug was dissolved into the sheets its circuits serve: the SWD + USART3 header, the rail probe header and the GND hooks all moved onto mcu, so SWDIO, SWCLK, SWO, MCU\_nRESET, DBG\_TX and DBG\_RX are sheet–local there and no longer cross a boundary. Test coverage lives on the circuit page it covers – docs/decisions/actuator–rev–testdebug.md.

GLOBAL POWER NETS CARRY NO SHEET PIN. GND, +5V, +5VA, +3V3 and +3V3A are Amodo power symbols and connect across the hierarchy on their own. power\_rails produces all four and owns their PWR\_FLAGS: power\_entry\_24v owns the flags for GND and +24V\_SW. Only +24V\_SW, V24\_LOGIC and V24\_MON leave power\_entry\_24v as ordinary nets, and they are drawn along the top of the sheet.

LAYOUT follows docs/FAFF–2–Electronics–Full.svg left to right: 24 V in at the top left, sensing and user I/O down the left column, the STM32H723VET6 in the middle, motor drive on the right, temp\_sense sits under motor\_drive rather than in the sensor column because it hangs off the same shared SPI2 CONFIG bus as the DRV8323 (ARCHITECTURE 5.1); the three bus wires are tapped once each, at the junctions between the two blocks. Its two probe channels stay inside its own sheet.

REFERENCE DESIGNATORS are allocated per sheet, 100 apart, from a fixed table – NOT from the page number, which changed when test\_debug went: 2xx power\_entry\_24v, 3xx power\_rails, 5xx loadcell\_afe, 6xx linear\_encoder, 7xx temp\_sense, 8xx nvm\_calibration, 9xx ui\_io, 10xx mcu, 11xx motor\_drive, 4xx is retired. Power symbols follow the same ranges (#PWR5xx, #PWR7xx, ...) so that no two sheets merge into one part at netlist time.

Requirements docs/REQUIREMENTS.md – architecture docs/ARCHITECTURE.md – decisions docs/DECISIONS.md and docs/decisions/actuator–sch–integrate.md – bring–up docs/TEST\_PLAN.md

Decisions: docs/DECISIONS.md

Requirements: docs/REQUIREMENTS.md

Hierarchical block map, wired; see docs/ARCHITECTURE.md

Proof–of–concept control board – development–board form factor

Amodo Design

Sheet: /

File: faff2\_cbs1.kicad\_sch

Title: ARIA\_SRB\_FAFF\_2 CBs\_1 – Root

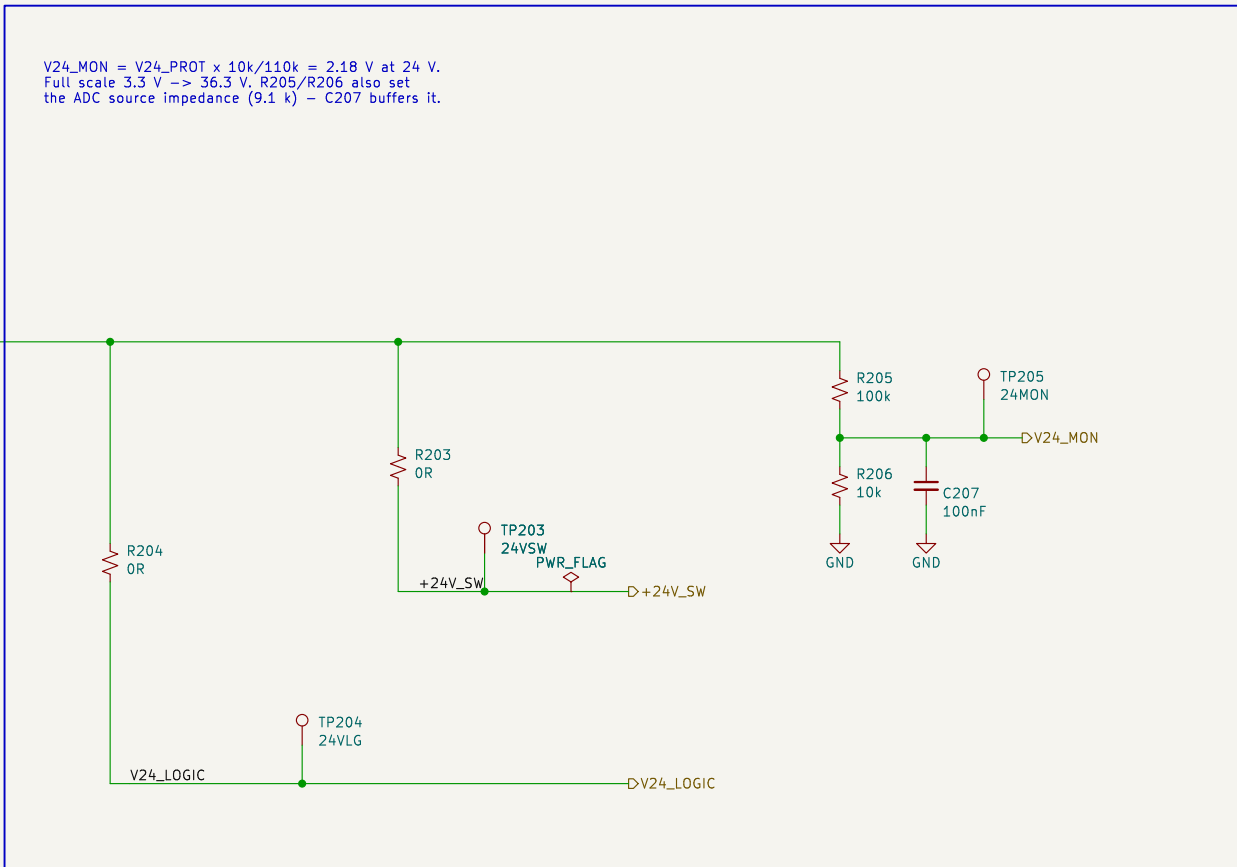
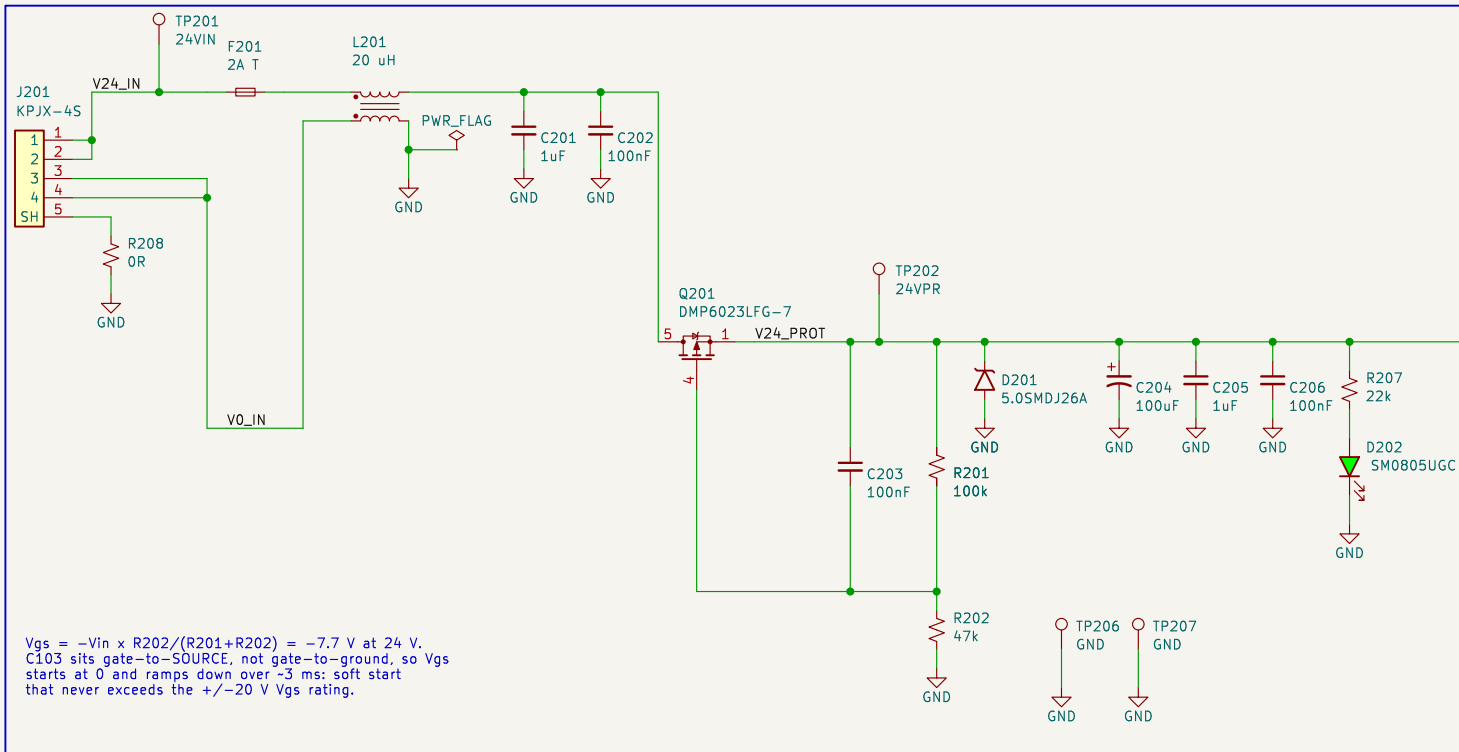
Size: A3

Date: 2026–09–03

Rev: 0.2

KiCad E.D.A. 9.0.8

Id: 1/10



BLOCK: power\_entry\_24v – 24 V DC input and protection (REQ-EL-01..04)

```
CHAIN J201.KPJX-45 (contacts doubled; 1,2 = +24 V, 3,4 = 0 V, 5 = shell)
-> F201 2 A time-lag (25 W peak = 1.04 A @ 24 V; 2 A T holds 1.5 A)
-> L201 common-mode choke, both lines; board GND is defined on its load side
-> Q201 P-channel reverse-polarity series switch (drain = supply, source = load,
so the body diode blocks a reversed supply and nothing downstream conducts)
-> D201 TVS + C204.C206 bulk/HF -> V24_PROT, the protected 24 V bus.
```

BRANCHES (docs/TES4\_PLAN.md 3.3 – each an isolation link AND a current break)  
 R203 -> +24V\_SW : motor\_drive only. Open R203 and the whole board can be exercised with no possibility of the actuator moving.  
 R204 -> V24\_LOGIC: power\_rails only. Independent of R203.  
 One OR link per branch serves as both the isolation link and the current break; a second series part would add joints and no capability.

GROUND One GND net board-wide. AGND / DGND / PGND in the block stub notes are layout partitions of it, not separate nets – see the decisions file.  
This sheet owns the single GND PWR\_FLAG; no other sheet may add one.

```

INTERFACES exported (hierarchical labels, wired to root sheet pins - DEC-0009)
  OUT +24V_SW -> motor_drive      OUT V24_LOGIC -> power_rails
  OUT V24_MON -> mcu PC1 / ADC1_INP11
  Global power nets used here: GND.

```

Reasoning, part sizing and residual ERC: docs/decisions/actuator-sch-power.md

Design reasoning: docs/decisions/actuator-sch-power.md  
Requirements: REQ-EL-01..04 - docs/REQUIREMENTS.md  
Independent motor / logic branch links (TEST\_PLAN 3.3)  
24 V in on KPJX-4S: fuse, CM choke, reverse polarity, TVS

**Amodo Design**

Sheet: /power\_entry\_24v/  
File: power\_entry\_24v.kicad\_sch

**Title: CBs\_1 – Power Entry 24 V**

|                    |                  |          |
|--------------------|------------------|----------|
| Size: A3           | Date: 2026-09-02 | Rev: 0.2 |
| KiCad E.D.A. 9.0.8 |                  | Id: 2/10 |

[illegible]

**B. POST-REGULATED 5 V RAILS (ADPL42005: +5V digital, +5VA analog)**

EN is tied to each LDO input – the datasheet's own automatic-startup connection – so the 5 V rails follow the pre-regulator with no sequencing logic. SENSE ties to VOUT at the regulator, not past the OR link, so lifting the link cannot open the loop. PG is unused; no-connect, not tied to RAIL\_PG00D. +5VA feeds ADS1235 / ADS1120 AVDD and the AFE excitation chain; +5V feeds the IKP11 read head.

The schematic diagram illustrates the LMR33630 buck converter circuit. The U304 LMR33630 IC is the central component, with its pins connected as follows:

- VIN (Pin 2):** Connected to the V24\_LOGICD input.
- EN (Pin 3):** Connected to the V24\_LOGICD input.
- PGND (Pin 1):** Connected to GND.
- VCC (Pin 6):** Connected to GND.
- TP (Pin 7):** Connected to GND.
- BOOT (Pin 7):** Connected to the SW pin (Pin 8).
- SW (Pin 8):** Connected to the SW pin (Pin 8).
- PG (Pin 4):** Connected to the SW pin (Pin 8).
- FB (Pin 5):** Connected to the FB pin (Pin 5).
- FB (Pin 5):** Connected to the FB pin (Pin 5).

The circuit includes several passive components:

- Resistors:** R308 (100k), R309 (8.2k), R310 (100k), R311 (43k), R312 (0R), R316 (1k).
- Capacitors:** C315 (10uF), C316 (10uF), C317 (220nF), C318 (1uF), C319 (100nF), C320 (22uF), C321 (22uF), C322 (100nF).
- Inductor:** L302 (6.8uH).

The output of the converter is connected to the +3V3 rail, which is also connected to the PWR\_FLAG pin (Pin 12) and the TP306 +3V3 test point.

FB301  
600R@100MHz

+3V3

R313  
100k

RAIL\_PG00D

+3V3A

C323  
10uF

C324  
100nF

+3V3

R314  
330R

D303  
SM0805URC

TP307  
+3V3A

TP308  
PGOOD

PWR\_FLAG

TP311  
GND

TP312  
GND

GND

GND

POWER SOURCE DECLARATIONS  
One PWR\_FLAG per rail produced here, at the regulator that makes it, on the load side of that rail's OR link. No other sheet may add one for these nets.

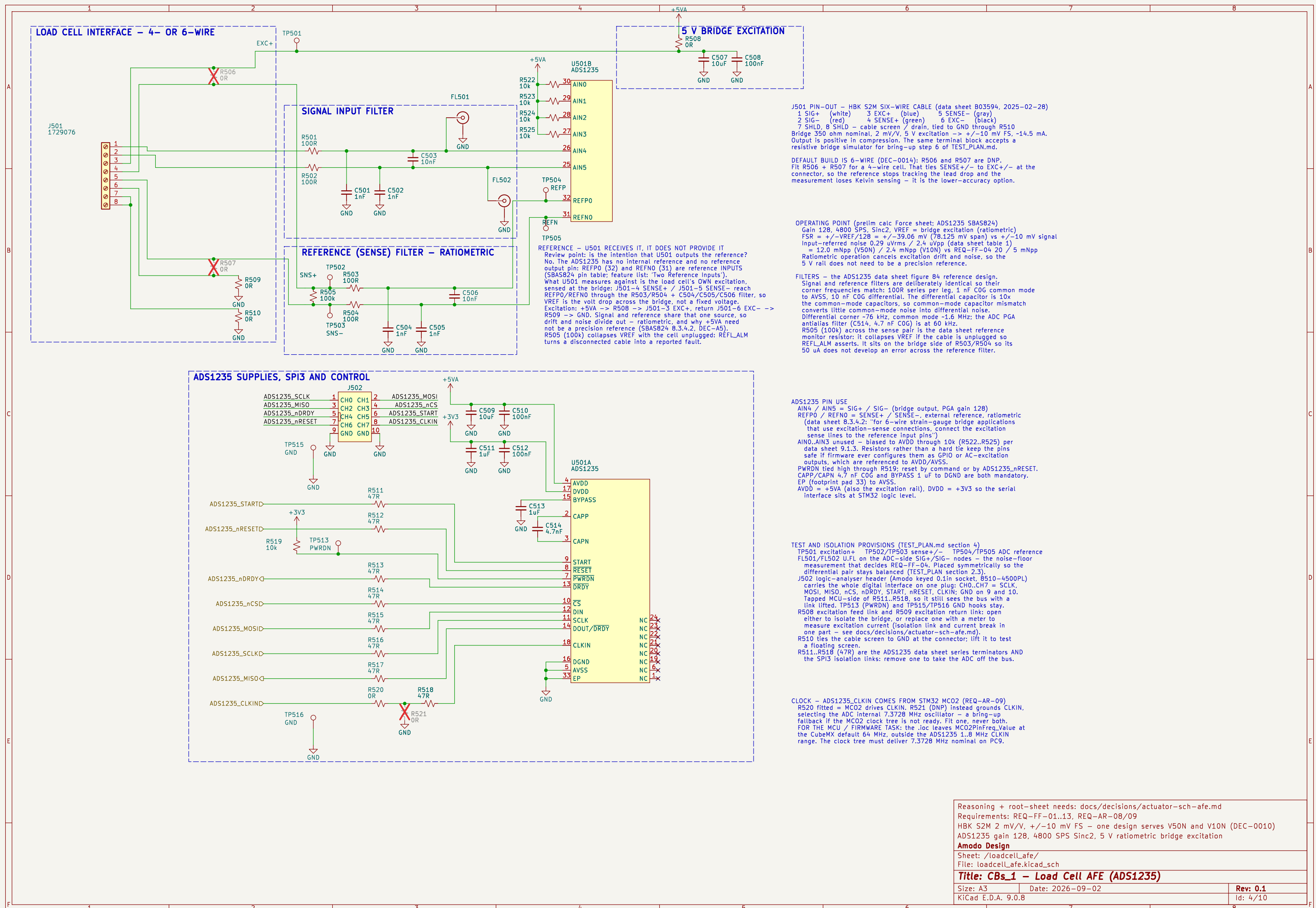
|       |       |                                                                                                                                                            |
|-------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RAILS | +5V5  | LMR33630 buck from V24_LOGIC. Pre-regulator only, local to this sheet.                                                                                     |
|       | +5V   | ADPL42005 LDO from +5V5. IKP11 read head (<65 mA plus RS-422 drive) and spare.                                                                             |
|       | +5VA  | ADPL42005 LDO from +5V5. ADS1235 / ADS1200 AVDD and the AFE 5 V excitation and reference chain, which lives in loadcell-Lafe - take it from +5VA, not +5V. |
|       | +3V3  | LMR33630 buck from V24_LOGIC. S1332, USB3320, DRV8323 logic, QSPI, EEPROM, RS-422 receiver, rotary encoder.                                                |
|       | +3V3A | -1.1 A worst case against a 3 A part.<br>ferrite-isolated from +3V3 for the ADC digital supplies.                                                          |

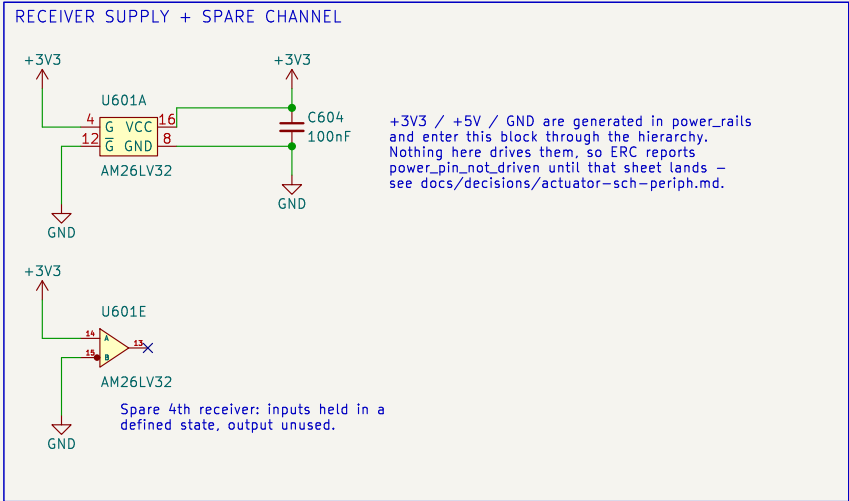
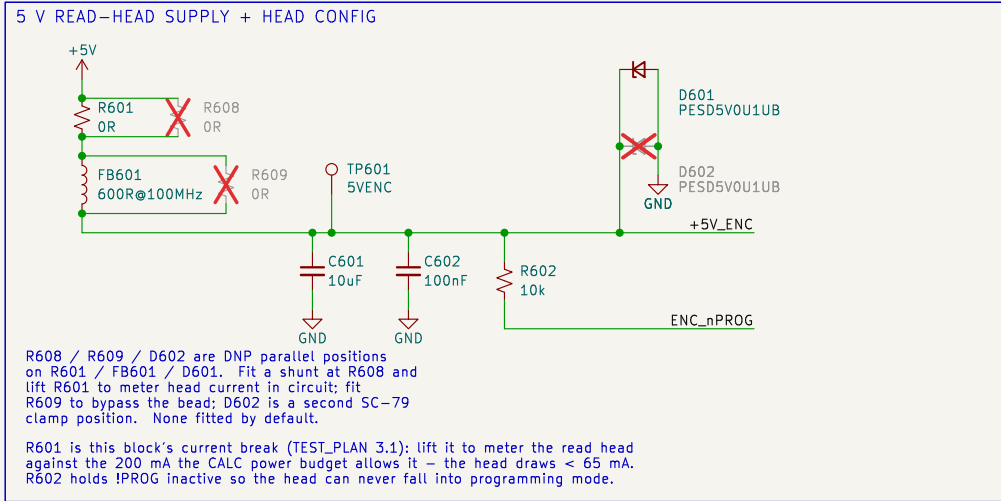
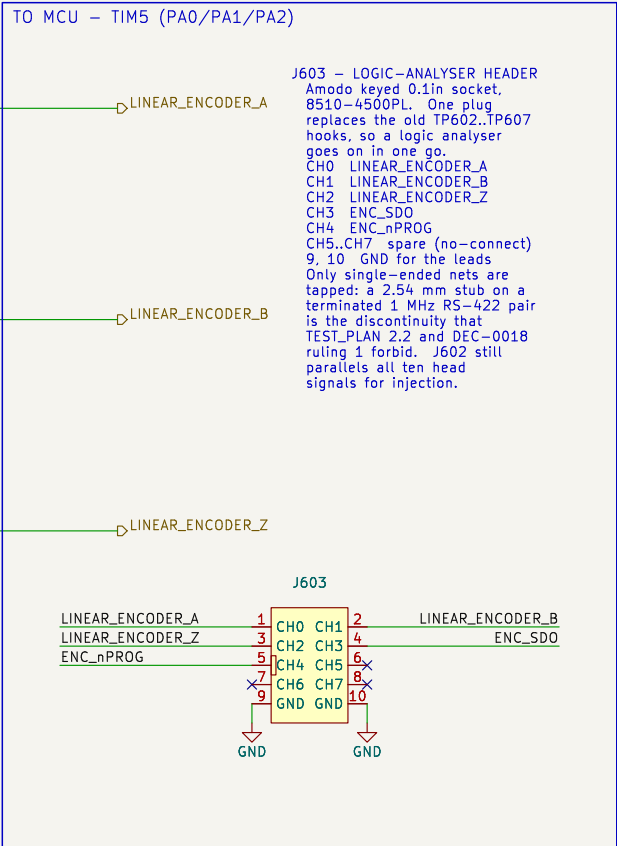
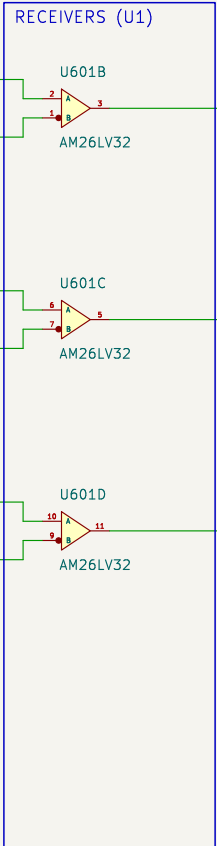
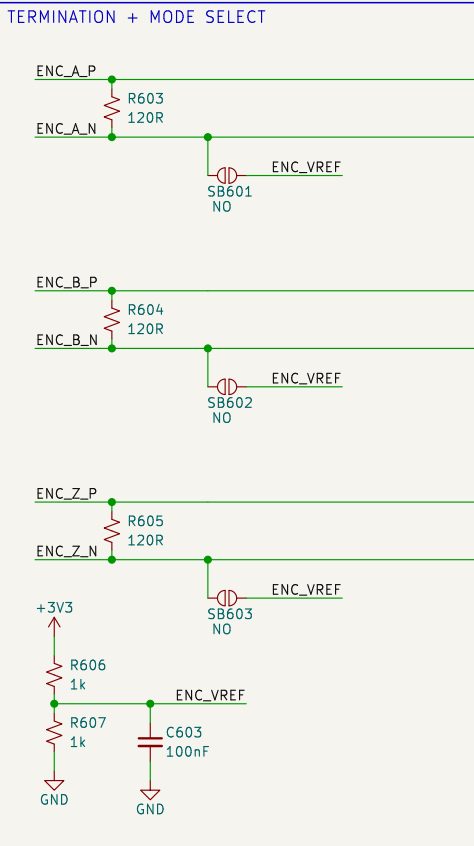
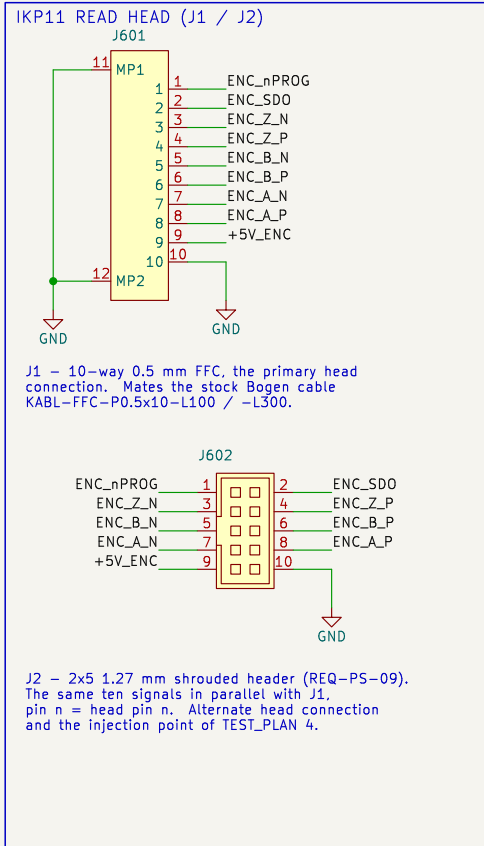
|   |   |
|---|---|
| 1 | 2 |
|---|---|

Reasoning, part sizing and residual ERC:  
docs/decisions/actuators-sch-power.md

POWER SOURCE DECLARATIONS  
One PWR\_FLAG per rail produced here, at the  
regulator that makes it, on the load side of  
that rail's OR link. No other sheet may add  
one for these nets.

|                                                            |                             |
|------------------------------------------------------------|-----------------------------|
| Design reasoning: docs/decisions/actuator-sch-power.md     |                             |
| Rail set and topology answer QQ-02 / QQ-07                 |                             |
| Per-rail isolation link / current break and test point     |                             |
| 24 V -> 5V5 pre-reg -> +5V and +5VA; 24 V -> +3V3 -> +3V3A |                             |
| <b>Amodo Design</b>                                        |                             |
| Sheet: /power_rails/                                       |                             |
| File: power_rails.kicad_sch                                |                             |
| <b>Title: CBs_1 – Power Rails</b>                          |                             |
| Size: A3                                                   | Date: 2026-09-02            |
| KiCad E.D.A. 9.0.8                                         | Rev: <b>0.2</b><br>Id: 3/10 |





#### BLOCK NOTES - linear\_encoder

Ordered head: Bogen IKP11-Z1.4-P1-V5-D1-R0.5-F1000-C1 (DEC-0003, REQ-PS-05)  
Z1.4 periodic index from the pole pitch, 4 counts P1 1 mm pole pitch  
V5 5 V supply, < 65 mA D1 RS-422 A/B/Z  
R0.5 0.5 um resolution, post-quadrature F1000 1 MHz per channel  
C1 15.8 x 15.4 PCB: a 10-way 0.5 mm FFC connector AND a 1.27 mm pad row

HEAD PIN ORDER (datasheet rev 3.7 p.6 - same order on the FFC and the pads)  
1 !PROG 2 SDO 3 /Z 4 Z 5 /B 6 B 7 /A 8 A 9 V+ 10 V-  
Check the FFC cable contact side (type A / type D) against J1 before ordering - J1 is a bottom-contact receptacle. J2 needs no such check and is the fallback.

#### MODES

RS-422, the ordered D1 build (default): R603/R604/R605 fitted, SB601/SB602/SB603 open.  
120 R across each pair, as the head datasheet asks (ZO = 120 R at the receiving end). No external fail-safe bias is needed: the AM26LV32 has open-circuit, shorted AND 100 R-terminated fail-safe, so an unplugged or dead head drives a static high - a stalled count, never phantom motion.  
Single-ended TTL / bench injection: remove R603/R604/R605, close SB601/SB602/SB603.  
Each inverting input then sits on ENC\_VREF (1.65 V), so a 0-3.3 V or 0-5 V single-ended source on ENC\_XP decodes correctly. This covers a TTL head variant (order code D3) and injection through J2 with no read head fitted.

U1 runs from 3V3, so its outputs are native 3V3 logic (REQ-PS-06); its -0.3 V to +5.5 V input common-mode range accepts the 5 V head directly, and it switches to 32 MHz against the 1 MHz per channel of REQ-PS-07. Both enables are tied active: G (pin 4) high, /G (pin 12) low.

No test points sit on the differential pairs: a probe stub on a terminated 1 MHz RS-422 line is a discontinuity (TEST\_PLAN 2.2), and the receiver outputs carry the same information single-ended. J603, the keyed logic-analyser socket, replaces the old TP602.TP607 hooks: CH0..CH4 = LINEAR\_ENCODER\_A / \_B / \_Z, ENC\_SDO, ENC\_nPROG, with GND on 9 and 10 for the ground leads. TP601 stays, a 1.0 mm SMT pad for a DMM on the 5 V head supply.

Test strategy: docs/TEST\_PLAN.md  
REQ-PS-01..10, REQ-AR-07, DEC-0003  
RS-422 A/B/Z -> 3V3 into TIM5 (encoder mode TI12 + CH3 capture)  
Bogen IKP11-Z1.4-P1-V5-D1-R0.5-F1000-C1 read head interface

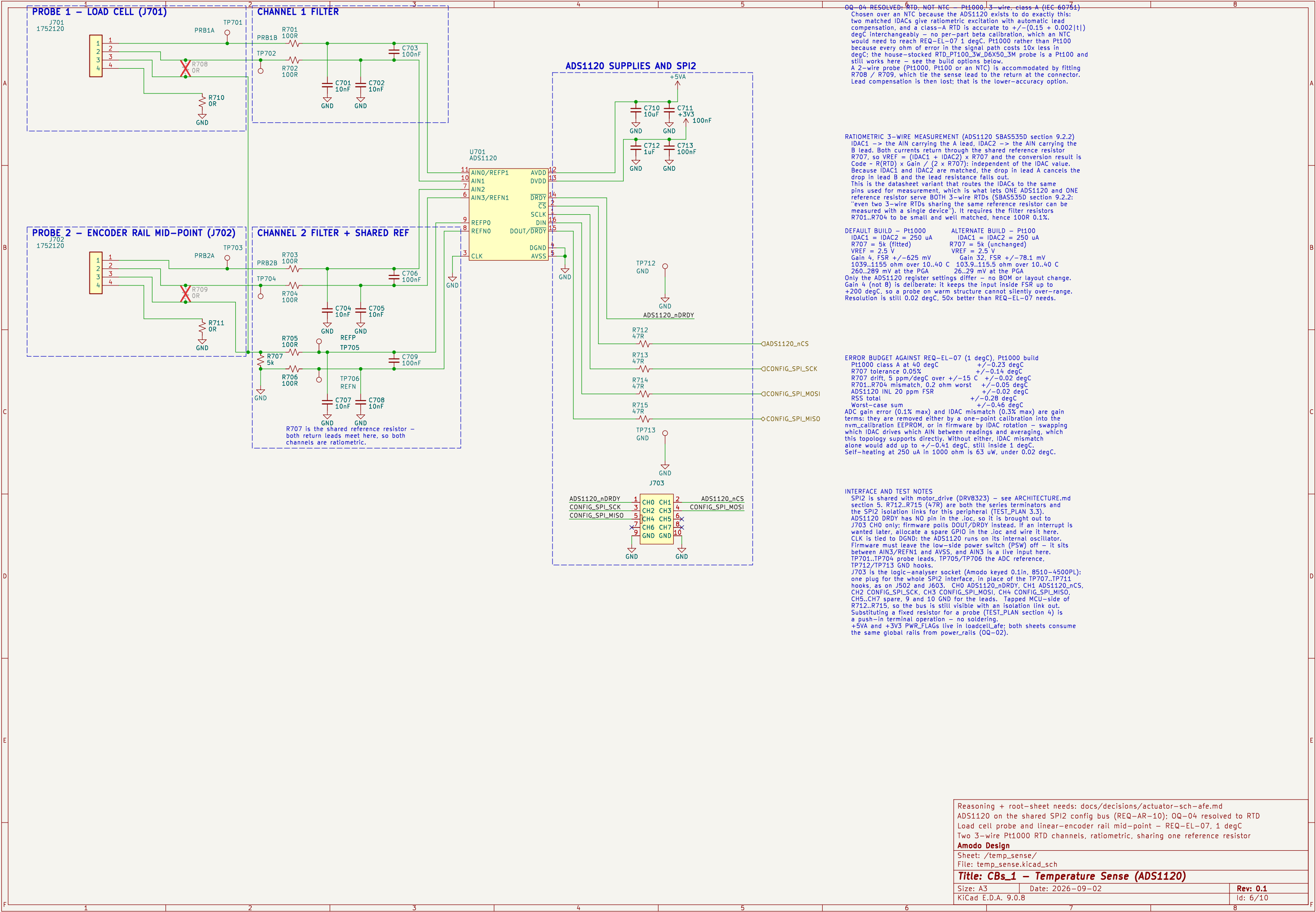
#### Amodo Design

Sheet: /linear\_encoder/  
File: linear\_encoder.kicad\_sch

#### Title: CBs\_1 - Linear Encoder (IKP11)

Size: A3 Date: 2026-09-02 Rev: 0.2  
KiCad E.D.A. 9.0.8 Id: 5/10





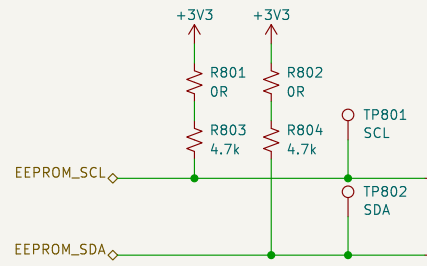
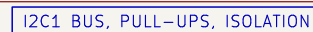
OQ-04 RESOLVED: RTD, NOT NTC - Pt1000, 3-wire, class A (IEC 60751)  
Chosen over an NTC because the ADS1120 exists to do exactly this:  
two matched IDACs give ratiometric excitation with automatic lead  
compensation, and a class-A RTD is accurate to +/- (0.15 + 0.002|t|)  
degC interchangeably - no per-part beta calibration, which an NTC  
would need to reach REQ-EL-07 1 degC. Pt1000 rather than Pt100  
because every ohm of error in the signal path costs 10x less in  
degC; the house-stocked RTD\_PT100\_3W\_D6X50\_3M probe is a Pt100 and  
still works here - see the build options below.  
A 2-wire probe (Pt1000, Pt100 or an NTC) is accommodated by fitting  
R708 / R709, which tie the sense lead to the return at the connector.  
Lead compensation is then lost; that is the lower-accuracy option.

RATIOMETRIC 3-WIRE MEASUREMENT (ADS1120 SBAS535D section 9.2.2)  
IDAC1 -> the AIN carrying the A lead, IDAC2 -> the AIN carrying the  
B lead. Both currents return through the shared reference resistor  
R707, so VREF = (IDAC1 + IDAC2) x R707 and the conversion result is  
Code - R(RTD) x Gain / (2 x R707); independent of the IDAC value.  
Because IDAC1 and IDAC2 are matched, the drop in lead A cancels the  
drop in lead B and the lead resistance falls out.  
This is the datasheet variant that routes the IDACs to the same  
pins used for measurement, which is what lets ONE ADS1120 and ONE  
reference resistor serve BOTH 3-wire RTDs (SBAS535D section 9.2.2:  
"even two 3-wire RTDs sharing the same reference resistor can be  
measured with a single device"). It requires the filter resistors  
R701..R704 to be small and well matched, hence 100R 0.1%.

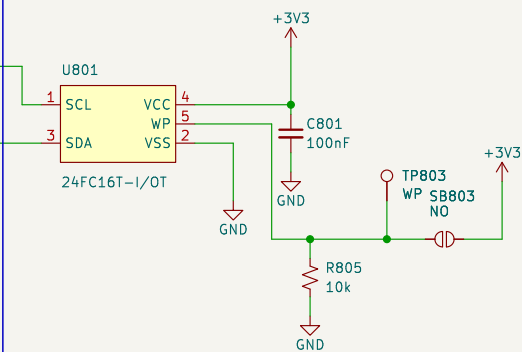
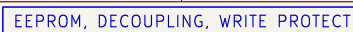
DEFAULT BUILD - Pt1000      ALTERNATE BUILD - Pt100  
IDAC1 = IDAC2 = 250 uA      IDAC1 = IDAC2 = 250 uA  
R707 = 5k (fitted)      R707 = 5k (unchanged)  
VREF = 2.5 V      VREF = 2.5 V  
Gain 4, FSR +/-625 mV      Gain 32, FSR +/-78.1 mV  
1039..1155 ohm over 10..40 C      103.9..115.5 ohm over 10..40 C  
260..289 mV at the PGA      26..29 mV at the PGA  
Only the ADS1120 register settings differ - no BOM or layout change.  
Gain 4 (not 8) is deliberate: it keeps the input inside FSR up to  
+200 degC, so a probe on warm structure cannot silently over-range.  
Resolution is still 0.02 degC, 50x better than REQ-EL-07 needs.

ERROR BUDGET AGAINST REQ-EL-07 (1 degC), Pt1000 build  
Pt1000 class A at 40 degC      +/-0.23 degC  
R707 tolerance 0.05%      +/-0.14 degC  
R707 drift, 5 ppm/degC over +/-15 C      +/-0.02 degC  
R701..R704 mismatch, 0.2 ohm worst      +/-0.05 degC  
ADS1120 INL 20 ppm FSR      +/-0.02 degC  
RSS total      +/-0.28 degC  
Worst-case sum      +/-0.46 degC  
ADC gain error (0.1% max) and IDAC mismatch (0.3% max) are gain  
terms: they are removed either by a one-point calibration into the  
nvm\_calibration EEPROM, or in firmware by IDAC rotation - swapping  
which IDAC drives which AIN between readings and averaging, which  
this topology supports directly. Without either, IDAC mismatch  
alone would add up to +/-0.41 degC, still inside 1 degC.  
Self-heating at 250 uA in 1000 ohm is 63 uW, under 0.02 degC.

INTERFACE AND TEST NOTES  
SPI2 is shared with motor\_drive (DRV8323) - see ARCHITECTURE.md  
section 5. R712..R715 (47R) are both the series terminators and  
the SPI2 isolation links for this peripheral (TEST\_PLAN 3.3).  
ADS1120 DRDY has NO pin in the .ioc, so it is brought out to  
J703 CH0 only; firmware polls DOUT/DRDY instead. If an interrupt is  
wanted later, allocate a spare GPIO in the .ioc and wire it here.  
CLK is tied to DGND; the ADS1120 runs on its internal oscillator.  
Firmware must leave the low-side power switch (PSW) off - it sits  
between AIN3/REFN1 and AVSS, and AIN3 is a live input here.  
TP701..TP704 probe leads, TP705/TP706 the ADC reference,  
TP712/TP713 GND hooks.  
J703 is the logic-analyser socket (Amodo keyed 0.1in, 8510-4500PL):  
one plug for the whole SPI2 interface, in place of the TP707..TP711  
hooks, as on J502 and J603. CH0 ADS1120\_nDRDY, CH1 ADS1120\_nCS,  
CH2 CONFIG\_SPI\_SCK, CH3 CONFIG\_SPI\_MOSI, CH4 CONFIG\_SPI\_MISO,  
CH5..CH7 spare, 9 and 10 GND for the leads. Tapped MCU-side of  
R712..R715, so the bus is still visible with an isolation link out.  
Substituting a fixed resistor for a probe (TEST\_PLAN section 4) is  
a push-in terminal operation - no soldering.  
+5VA and +3V3 PWR\_FLAGS live in loadcellLafe; both sheets consume  
the same global rails from power\_rails (OQ-02).

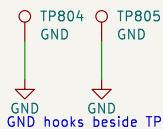


R801/R802 are the removable links of TEST\_PLAN 4: lift either to drop its pull-up. SB801/SB802 are cut to take the EEPROM off I2C1 while the bus stays pulled up – this block owns that break, so mcu must not duplicate it. The SDA pull-up crosses the SCL run, no junction.



WP is ACTIVE HIGH. R805 holds it low, so the part is writable and firmware can store calibration at any time; solder SB803 to lock the device permanently. TP3 lets a jig drive WP either way. This resolves OQ-06 – no MCU pin is spent on write protect.

## GROUND HOOKS



+3V3 and GND are generated in power\_rails and enter this block through the hierarchy; nothing here drives them, so ERC reports power\_pin\_not\_driven until that sheet lands. See docs/decisions/actuator-sch-periph.md.

GND hooks beside TP801/TP802 for the scope or analyser clips.

## BLOCK NOTES – nvm\_calibration

U1 = Microchip 24FC16T-I/OT, 16 kbit (2048 x 8) I2C EEPROM in SOT-23-5.

Pin map verified against the datasheet package drawing (DS20001703R p.1):

1 SCL 2 VSS 3 SDA 4 VCC 5 WP.  
1 7-5.5 V and 1 MHz capable, so it keeps

1.7–5.5 V and 1 MHz capable, so it keeps up with I2C1 at either 400 kHz or Fast-mode Plus, and 4 million erase/write cycles covers repeated calibration.

2 KB is sized for coefficients, not tables: force and temperature calibration constants, the variant (V50N / V10N – see DEC-0010, where the variants differ only by a calibration constant), board serial number and build state. Bulk force profiles live in the OCTOSP11 RAM on the mcu sheet (DEC-0006), not here.

**ADDRESSING CAVEAT.** The 24xx16 uses all three block-select bits internally, so it answers to the **WHOLE** 1010xxx address space and A0/A1/A2 do not exist on the part. No second 1010-family device can share I2C1. It is the only device on the bus today; if another is ever added, swap to a smaller 24xx02/04 first.

4k7 pull-ups suit 400 kHz on a short board-level bus with one load. They are the only pull-ups on I2C1: the mcu block must not add its own.

Test strategy: docs/TEST\_PLAN.md  
REQ-EL-08, REQ-AR-11; resolves OQ-06 (write protect)  
I2C1: EEPROM\_SCL PB8, EEPROM\_SDA PB9  
16 kbit I2C EEPROM for calibration and compensation data

## Amodo Design

Sheet: /nvm\_calibration/  
File: nvm\_calibration.kicad\_sch

**Title: CBs\_1 – NVM / Calibration Store**

Size: A3

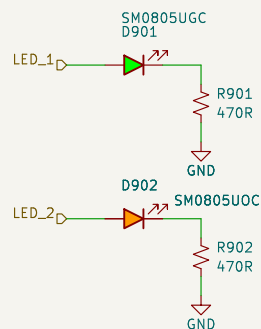
Date: 2026-09-02

Rev: 0.2

KiCad E.D.A. 9.0.8

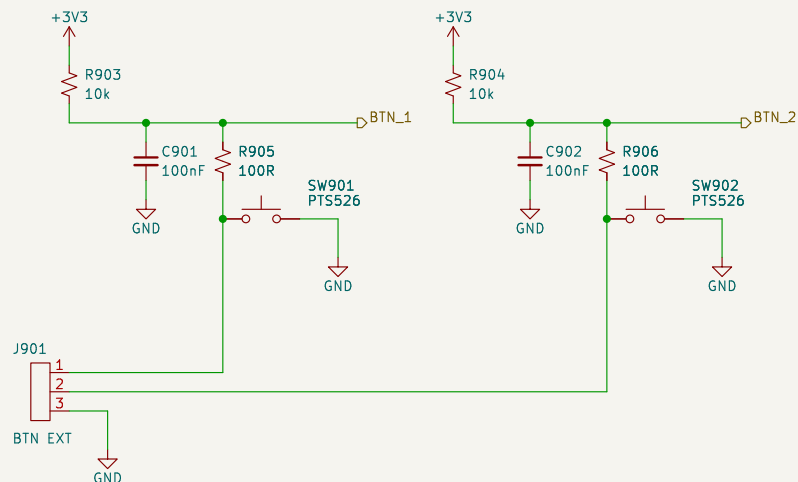
Id: 7/10

## STATUS LEDS



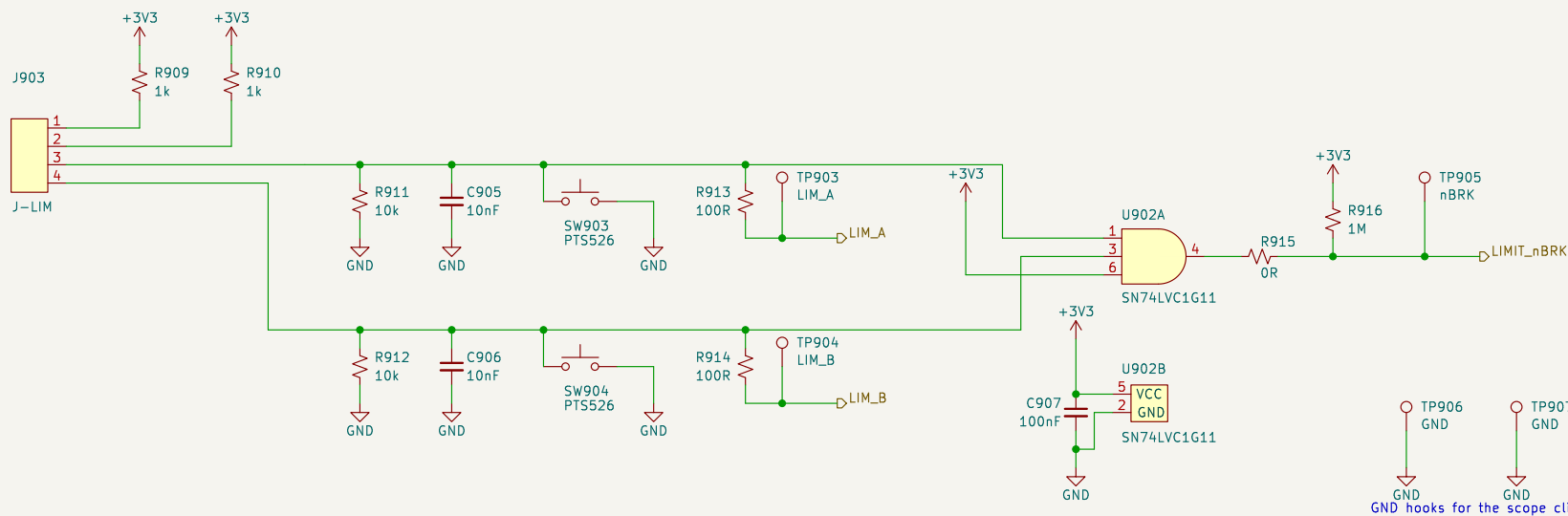
Active high, about 2.8 mA each.  
LED\_1 green is the heartbeat of  
TEST\_PLAN bring-up step 3; LED\_2  
orange is the fault / status lamp.

USER BUTTONS (BTN\_1 = HOME)



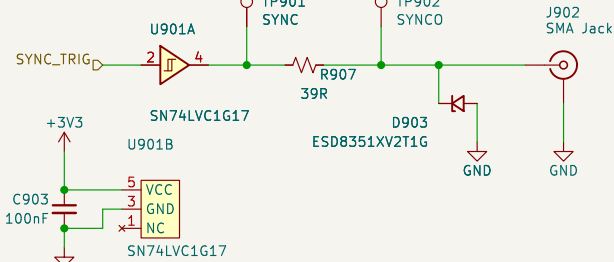
REQ-CC-06 asks for homing from the API OR an external button, so J901 parallels both on-board switches for a panel button. R905/R906 limit the contact current and set the debounce with C901/C902 – about 1 ms rise, 10 us fall. Which of BTN\_1 / BTN\_2 is HOME is OQ-05: firmware and silkscreen only, the two channels are electrically identical.

## END-OF-STROKE LIMIT SWITCHES + TIM1 BREAK



J-LIM pinout: 1 LIM\_A feed 2 LIM\_B feed 3 LIM\_A return 4 LIM\_B return.  
Both switches are NORMALLY CLOSED, so a closed switch pulls its node HIGH (3.0 V) and an open switch – or a broken wire, or an unplugged connector – lets the 10 k pull-down take it LOW. LOW therefore means ‘at that end of stroke, or the wiring has failed’, which is the safe direction to fail in.

SYNC / TRIGGER OUT (TIM15\_CH1)



REQ-EL-05 / REQ-EL-06: 3.3 V push-pull, 50 ohm source terminated, SMA jack. U901 buffers PE5 so the MCU pin never sees the cable and the source impedance is repeatable: R907 39 R plus the LVC output impedance (10–15 ohm typical) is about 50 ohm. TP901 is pre-termination, TP902 post; the SMA itself is the coaxial access of TEST\_PLAN 2.3. D903 is 0.55 pF and does not soften the edge.

BLOCK NOTES - ui\_io

SAFETY. SPEC 9.1 asks only for a firmware kill on limit actuation (REQ-SF-002) and a block diagram and the spec both carry a second, independent hardware path (REQ-SF-05, DEC-0012): U002 ANDs the two healthy-limit nodes, so LIMIT\_nBRK is HIGH only while both switches are closed and both harness legs are intact. Any limit reached, any broken wire, an unplugged J-LIM or a dead 3V3 rail drives it LOW into TIM1\_BKIN2, and the PWM outputs go inactive with firmware taking no part. LIM\_A / LIM\_B reach the MCU on their own EXTL pins for the firmware kill and for telling the two ends apart. The AND gate's third input is tied high.

R915 is the TEST\_PLAN 3.3 link that lets the two TIM1 BREAK sources be exercised alone. It is fitted by default and must be marked on the silkscreen. With it lifted, R916 (1 M) holds LIMIT\_NBRK high so the DRV8323\_nFAULT path can be tested on its own; the firmware kill through LIM\_A / LIM\_B is unaffected either way. 1 M is weak enough that a tripped chain still pulls the net down to about 30 mV.

SW903 / SW904 are the 'assert each limit without the mechanics' means of TEST\_PLAN 4: pressing one pulls its node low through the 1 k feed (3.3 mA), which is exactly the state that end of stroke produces. They act independently because the two switch loops are independent – a series safety chain could not be stimulated one end at a time.

C905 / C906 (10 nF) with the -0.9 k source give about 9 µs of rise filtering and 100 us of fall; at the 20 mm/s of REQ-ME-03 that is 2 um of extra travel before the break asserts, against millimetres of switch overtravel. R913 / R914 protect the MCU pins. No discrete TVS is fitted on the limit lines: the harness stays inside the instrument and the RC plus the series resistor already bound the current into the MCU clamps. The SMA is the one externally exposed pin here, and it does get a clamp (D903).

+3V3 and GND are generated in power\_rails and enter this block through the hierarchy; nothing here drives them, and power\_rails owns their PWR\_FLAGS, so project ERC is clean at severity=all.

Test strategy: docs/TEST\_PLAN.md  
 REQ-EL-05/06/09, REQ-CC-06, REQ-SF-01/02/05  
 LIMIT\_NBRK PE6 -> TIM1\_BKIN2: hardware motor kill (DEC-0012)  
 Status LEDs, user buttons, SMA SYNC output, end-of-stroke limits  
**Amodo Design**  
 Sheet: /ui\_io/  
 File: ui\_io.kicad\_sch

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**Title: CbS\_1 – User I/O, SYNC, Limit Switches**

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